

Sensor Averaging

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1 Introduction

Averaging multiple sensor readings of zero bias sensors reduces error, for example in GPS, if we connect to multiple GPS servers, and average the locations given, we can reduce error in position prediction.

X ground truth position of device

$\hat{X}_i$ position predicted from satellite i

$$E[\hat{X}_i - X] = 0 \quad (0 \text{ bias for simple example})$$

$$E[(\hat{X}_i - X)^2] = error_i \quad (variance_i)$$

$$\text{theorem 1:} \quad E[(\sum \hat{X}_i)/n - X] = 0$$

$$\text{theorem 2:} \quad E[(\sum \hat{X}_i)/n - X]^2 < error_i$$

2 Theorem 1

$$E[(\sum \hat{X}_i)/n - X] =$$

$$E[(\sum^n \hat{X}_i)/n - nX/n] =$$

$$E[(\sum^n \hat{X}_i)/n - (\sum^n X)/n] =$$

$$E[1/n(\sum^n \hat{X}_i - \sum^n X)] =$$

$$E[1/n \sum^n (\hat{X}_i - X)] =$$

$$1/n \sum^n E[\hat{X}_i - X] = 1/n \cdot n \cdot 0 = 0$$

3 Theorem 2

$$E[(\sum \hat{X}_i)/n - X]^2 < error_i$$

$$E[(\sum \hat{X}_i)/n - X]^2 =$$

$$E[(\sum \hat{X}_i)/n - nX/n]^2 =$$

$$E[(\sum \hat{X}_i)/n - (\sum X)/n]^2 =$$

$$1/n^2 E[(\sum \hat{X}_i) - (\sum X)]^2 =$$

$$1/n^2 E[(\sum (\hat{X}_i - X))]^2 =$$

$$1/n^2 E[(\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \dots + (\hat{X}_n - X)]^2 =$$

$$\begin{aligned} 1/n^2 E[& (\hat{X}_1 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \dots + (\hat{X}_n - X)) + \\ & (\hat{X}_2 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \dots + (\hat{X}_n - X)) + \\ & (\hat{X}_3 - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \dots + (\hat{X}_n - X)) + \\ & \vdots \\ & (\hat{X}_n - X)((\hat{X}_1 - X) + (\hat{X}_2 - X) + (\hat{X}_3 - X) + \dots + (\hat{X}_n - X))] = \end{aligned}$$

$$1/n^2 E[((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X)) n] =$$

$$1/n E[((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X))] =$$

Now we need the following lemma:

X_i and X_j independent gives $E[(\hat{X}_i - X)(\hat{X}_j - X)] = 0$

proof:

$$E[(\hat{X}_i - X)(\hat{X}_j - X)] =$$

$$E[(\hat{X}_i \hat{X}_j - \hat{X}_i X - X \hat{X}_j + X^2)] =$$

$$\begin{aligned}
E[\hat{X}_i \hat{X}_j] &= E[\hat{X}_i]X + XE[\hat{X}_j] - X^2 = \\
&(\hat{X}_i, \hat{X}_j \text{ independent}) \\
E[\hat{X}_i]E[\hat{X}_j] &= E[\hat{X}_i]X + XE[\hat{X}_j] - X^2 = \\
X \cdot X &= X \cdot X - X \cdot X = X^2 = 0 \\
(E[\hat{X}_i] = X \text{ for simplicity we assume zero bias as stated above.})
\end{aligned}$$

Now, continuing with the proof using the lemma $E[(\hat{X}_i - X)(\hat{X}_j - X)] = 0$

$$\begin{aligned}
1/n \ E[((\hat{X}_i - X)^2 + (n-1)(\hat{X}_i - X)(\hat{X}_j - X))] &= \\
1/n \ E[((\hat{X}_i - X)^2)] &= error_i = E[(\hat{X}_i - X)^2]
\end{aligned}$$

This is nice, as it shows the expected error of n sensors averaged is $1/n$ times the single sensor error.